

Problem Statement

"DataFlow Inc., a growing data-driven organization, is struggling to answer basic data-related questions. Their data is scattered across raw CSV and JSON files — transactions, logs, and product/customer data — with no central place to analyze it. The team cannot detect schema changes, data drift, or ensure data quality. The data engineering team has been asked to build a centralized platform that ingests raw data, cleans it, detects changes, and maintains historical data for reliable analytics."

Tech stack used:

Python Basics Load CSV/JSON files

Data Structures Handle data

Python Libraries Pandas

Data Processing Cleaning

SELECT + Keys Basic selection

WHERE + Indexes Filtering

GROUP BY Aggregations

JOINS + Foreign keys Data comparison

CASE + ACID Conditional logic

CTE Transformations

Window Functions Trend analysis

ETL Concepts Data ingestion

ADF Pipelines Orchestration

Spark Basics Distributed processing

DataFrames Structured data

Read/Write CSV/JSON

Transformations Data cleaning

Delta Lake Versioning

Data Modeling Structuring

SCD + Merge History tracking

Medallion Bronze/Silver/Gold